The Phenomenon of the World ⇒ A Redefinition of Space

Working Paper

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Abstract

This paper proposes a redefinition of space that follows from the phenomenology of the world as a boundary event. Space is not treated as a container preceding entities; it is construed as a secondary order generated by entities, by the rules of their existence, and by the network of relations among them. The key move is an inversion of the customary sequence: not "space → rules → objects," but "entities and rules → network of relations → local region → space as the form of its presentation." Two registers are distinguished, whose conflation generates the principal difficulties of interpretation: space-for-an-observer (extension as a mode of givenness) and space-as-topology (the network of permissions, prohibitions, and invariants of connectivity). It is shown that "place" is an effect of occupancy rather than a pre-existing volume; that emptiness is not encountered in any operation; that geometry is a trace of the configuration of rules rather than its condition. The asymmetry of scale is introduced as a working constraint on any localization. The paper formulates a protocol of analysis and the conditions under which the proposed treatment would be refuted.

Keywords: space, container model, inversion, placeholder, boundary, topology of relations, asymmetry of scale, gravity, operational phenomenology.

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1 The Problem

The concept of space is used as self-evident: a stage on which everything else is arranged. This self-evidence is not the result of verification. It is a habit of representation inherited together with the language of description.

The task of the paper is formulated narrowly: **to describe localization, distance, and**

extension without invoking the supposition of a container that cannot be detected in any operation.

The test of the initial supposition is simple and requires no apparatus. Let us attempt to point to space itself—in a point, in a region, in a gap. Each time, entities are encountered: a stone, air, an atom, a field, a wave—or else the rules by which they exist. The same occurs in a thought experiment: it operates with the same entities and rules, only under different assumptions. The empty container does not figure in any protocol—neither instrumental nor mental.

It does not follow from this that "space does not exist." What follows is more precise: **it does not occupy the place in the order of explanation that container intuition ascribes to it.**

2 The Inversion

The container sequence runs: first the arena, then the rules of its marking, then the objects inhabiting it.

The proposed sequence is the reverse. An entity exists according to certain rules—this is not a definition by genus and species, but a necessary tautology: to exist means to exist somehow. Rules determine what can be combined with what, what is forbidden to what, what follows from what. The aggregate of these permissions and prohibitions forms the network of relations in a local region. The description of this region is what we call space.

The inversion is not a rearrangement of words. It changes the object of inquiry. Instead of the question "what is space like inside an apple, an atom, a wave," we ask: "what rules operate in crossing the boundary of this apple, atom, wave?" The first question, upon crossing boundaries, generates accumulating difficulties of interpretation: one has to imagine a container inside that which itself occupies a place, and then a container inside a container. The second question generates no difficulties of this order: it requires only that the change of rules be described.

Space is not a condition of the existence of entities. On the contrary, entities, actualizing their properties and building networks of connection, generate space as the local geometry of their relations. What we perceive as "volume" or "emptiness" is nothing other than the sensation of low density of ontological connections in a given regime. *Here, "regime" denotes not a substantial domain of being but an operational condition of distinction—a stable aggregate of rules, parameters, and conditions under which a system functions, is observed, or manifests its properties (a strict analogue of "frame of reference" in physics or "paradigm" in the methodology of science).*

The same move works outside the physical world. A number does not occupy the place of another number; an axiom does not combine with a theorem; in the world of abstractions, "place" is likewise a network of relations and prohibitions rather than a volume. This shows that what is at issue is not physical extension but the structure of distinction as such.

3 The Placeholder: Place as an Effect of Occupancy

One object cannot occupy the place of another. This prohibition is usually explained by the fact that the place already existed and the object occupied it. But operationally what is registered is the reverse: **place is recognized only through occupancy.** A placeholder is not a cell awaiting content; it is a zone delineated by the rules of non-combination of a given entity with others.

In physics this manifests as a prohibition principle (for example, the Pauli principle for fermions); in the world of abstractions, as the uniqueness of designation. A prohibition principle is not a property of space but a property of regime: the rules of a given ontological context do not permit superposition.

Two consequences follow.

- First: the prohibition of combination is prior to extension. What we call extension is what is composed of the aggregate of such prohibitions and permissions.
- Second: the sensation of "volume" and "emptiness" is not a reflection of the presence or absence of a container. It is the form in which a given density of relations is given to an observer.

4 Two Registers

The conflation of two registers is the principal source of difficulties. They should be distinguished directly.

Space-for-an-observer. Extension, direction, "here" and "there," the sensation of an empty gap. This is the mode of givenness dependent on the regime in which the observer works: on the observer's range of distinction, on the channels of connection with the world, on which entities are manifested for the observer at all. The register is not illusory: it is exact within its boundaries and is the sole ground on which intuition is possible.

Space-as-topology. The network of permissions, prohibitions, and invariants of connectivity. Here there is no volume, no "empty gap," no directions in the customary sense. There is what is connected with what, under what conditions a connection is actualized, what is preserved under transformations.

These are not two objects and not two layers of reality. They are two readings of one network. Neither reduces to the other: topology does not explain why extension feels precisely as it does; the sensation of extension does not give access to invariants.

A practical rule: **the intuition of the first register is not transferred into the second without loss of correctness.** Most "paradoxes of space" arise precisely from such transfer—when the properties of the mode of givenness are ascribed to the structure of relations.

5 Distinguishing an Environment without Contact

A useful test situation. An object does not touch other objects. Does it nonetheless distinguish its environment?

A stone heated on one side "presupposes" a source of heat. The word "presupposes" here is not anthropomorphic: what is at issue is that an asymmetry of state is a trace of an asymmetry of connections, not a result of contact. The question about falling is posed analogously: is falling in a tube without touching the walls distinguishable from falling outside a tube, and does the width of the tube matter?

The treatment in the proposed terms is as follows: the absence of contact is not the absence of connections. What matters is not the "free volume around" but the configuration of rules in which the object is included. The width of the tube is of interest not as the size of a volume but as a parameter of this configuration.

What matters here is **epistemic honesty regarding its status**. This is a **formulation of the problem, not an assertion of a result.** The paper does not decide whether a difference will be detected and in which regime; it indicates that the question should be formulated through the configuration of rules rather than through the volume of a container. If no difference is detected where the configuration of rules is manifestly different, the proposed treatment will be weakened.

6 The World as Boundary: The Chess Model

The model is convenient in that everything in it is explicitly given and nothing has to be completed.

The pieces have type-names, but each piece in a game is unique: it has its own position, its own history, its own fate. Its square cannot be occupied by another piece simultaneously with it. At the same time, the situations for one's own and the opponent's colour are fundamentally different: one's own piece cannot occupy this square at all; another's can—by removing the occupant. The prohibition is thus not geometrical but relational: it depends on who the other entity is.

A piece removed from the board loses part of its uniqueness—not material but situational. Its determinacy was the determinacy of a position in a network. Outside the network it is again merely a **generic unit**. This is a direct illustration of the thesis: determinacy is a trace of inclusion, not an internal property.

Empty squares appear to be ready **volumes**—exemplary "containers." But a pawn does not move backward, and the squares behind it do not exist for it as reachable. An empty square is a placeholder and a node of rules, not a container. The "pure emptiness" of the board is an abstraction whose status is that of **pure potentiality** only.

Hence the general conclusion: **the boundary of a world does not run along the edge of a region; it runs along the limit of permissions.** A world is determined not by what it includes but by what is permitted in it.

The chess model is not a simple analogy. It is a structural isomorphism: the prohibition of

occupying a square with one's own piece and the Pauli prohibition principle are one and the same operator (the prohibition of combination), realized in different regimes. The model exhibits a structure; it does not illustrate it.

7 Geometry without a Container

There exists a radical formulation: space as an independent entity does not exist; there is only geometry; calculations should be conducted without regard to intuition. Formally this works. Psychologically and methodologically it leaves the investigator without ground: intuition is declared an obstacle, and its testimony is held to be inadmissible.

The proposed inversion relieves this tension **elegantly**. Geometry is the description of rules and their invariants, not the deformation of an invisible fabric. The intuition of extension is not erroneous—it is regime-local and exact within its regime. It ceases to be an obstacle and becomes the testimony of a concrete register that has boundaries of applicability.

The upshot is this: geometry is a trace of configuration, not its condition. Intuition returns as a legitimate source of data about the regime—with a mandatory indication of its boundaries.

8 The Asymmetry of Scale

The asymmetry is introduced as a working constraint, not as an elegant symmetrical scheme.

In moving "inward," toward a decrease in the scale of distinction, a rigid limit is reached beyond which distinction ceases to be determinate: the rules of the given regime are exhausted, and what follows is not a finer detail of the same order but a different order of description. In moving "outward," toward enlargement and complexification, no analogous limit is discovered: any attained level is included in the next.

The consequences for localization are direct. The concepts of "place" and "distance" are defined in the middle part of the range and lose determinacy at its edges. This is not a gap in knowledge that will be closed by an increase in precision; it is a property of the very connection between a regime and its content.

Hence a working hypothesis about the nature of difficulties at extreme scales: the divergence of descriptions at different scales is a divergence of regimes of distinction, not a defect of **the** mathematical apparatus. Accordingly, the productive question is not "how to unify the descriptions" but "how the transition between them is structured." The hypothesis is marked as a hypothesis: it indicates a direction of search, not a ready-made unification.

9 Heuristic Vector: Physics and the Boundaries of Regimes

The proposed inversion allows the conflict between General Relativity and Quantum Mechanics to be reinterpreted not as a mathematical incompatibility but as a collision of two different ontological regimes of distinction, each of which has its own rigid lower limit. The paper does not assert that this is so; it asserts that this is a productive way of posing the question: instead of a

search for a "theory of everything"—a search for an operator of translation between incompatible languages of description. **From this perspective**, gravity and geometry are reinterpreted not as the deformation of an invisible fabric but as the manifestation of the rules of translation at the boundary of regimes.

How exactly can the meta-language help the physicist? It does not supply a ready-made hypothesis, but it changes the form of the question. Instead of "how to unify GR and QM"—"where does the boundary of the regime pass at which geometric extension loses determinacy?" Instead of "what is the nature of the singularity"—"what happens to the rules of distinction at the limit of scale?" These reformulations do not solve the problem, but they make it accessible to operational analysis: one can search not for an equation but for the **point of regime change** and its parameters. This is already productive: the physicist receives not an answer but a coordinate in which the answer should be sought.

10 The Protocol of Analysis

For any phenomenon in which the question of space arises, the following five-step sequence is proposed.

1. First: to indicate the entities and the rules by which they exist—without presupposing an arena in advance.
2. Second: to identify the prohibitions and permissions of combination; thereby to obtain placeholders rather than **spatial containers**.
3. Third: to determine what exactly changes in crossing the boundary of the region under consideration—the composition of entities, the rules, or both.
4. Fourth: to check the asymmetry of scale: where the lower limit of distinguishability lies and what happens upon enlargement.
5. Fifth: to distinguish the registers—to separate the description of the mode of givenness from the description of the topology of relations, and not to transfer the intuition of the first into the second.

The protocol requires the adoption of no metaphysics. It requires only the rejection of an undescribed supposition of a container.

11 What the Paper Does Not Assert

The paper does not propose a new physical theory, does not derive equations, does not abolish metric descriptions, and does not claim to predict numerical values. It proposes an order of explanation and a language in which this order is observed.

The conditions of refutation are formulated directly. The treatment would be refuted if: an operation were discovered that fixes space without entities and without rules; a system were shown to preserve the regime of distinction under unbounded decrease of scale; or a consistent and

complete localization without asymmetry of limits were constructed. None of these conditions is at present satisfied; their satisfaction would render the proposed inversion untenable.

The limit of the paper itself is noted separately: it is conducted from within a regime of distinction and therefore submits to the same constraint that it describes. Its definitions have a solid core and an open periphery. This is not a concession to criticism but a condition of its correctness.

12 Result

Space is not a stage. It is a result: a secondary order generated by entities, by the rules of their existence, and by the network of their relations. Place is an effect of occupancy; emptiness is not encountered in any operation; geometry is a trace of the configuration of rules; extension is the mode of givenness of this configuration in a concrete regime, with determinate boundaries of applicability.

The inversion removes the difficulties that arise in attempting to imagine a container inside that which itself occupies place, and restores to intuition the status of legitimate testimony—with an indication of its limit.

The question shifts: not "what is located in space" but "what rules operate and where the boundary passes beyond which they cease to work."

Appendix: FAQ for a Potential Reviewer

Q1. Is this not simply a paraphrase of the relational ontology of Leibniz or Mach?

A. Partly—yes, the historical parallels are obvious. But the difference is that we do not merely assert "space is the order of things"; we give an operational protocol: how exactly this order is generated, which properties participate in the act of fixation, where the boundary between regimes passes. This is not philosophy but a meta-language with a procedure.

Q2. Is this physics or philosophy?

A. Neither, in the strict sense. It is a meta-language for physics—not replacing it but giving it a way to speak about **what it leaves unspoken**: the rules under which something becomes available for description at all. Physics describes what happens within a regime; the paper describes how regimes replace one another and where their boundaries pass. It does not derive equations and does not predict numbers; it indicates which question should be posed for an equation to become possible at all.

Q3. Are there not too many terms—"ontological regime," "placeholder," "mode of givenness"?

A. Each term is introduced for the sake of a distinction. "Regime"—to speak of incompatible languages of description. "Placeholder"—to speak of place as an effect of occupancy rather than as a volume. "Mode of givenness"—to separate extension as givenness from the topology of relations.

If any of them is removed, the corresponding distinction can no longer be drawn. The terminology is functional, not decorative.

Q4. Does this not contradict Einstein?

A. On the contrary, it develops his intuition. Einstein said: "space does not exist; there is only geometry." We say: "geometry is a trace of the configuration of rules, not its condition." This is not an abolition but a deepening: geometry is not abolished but ceases to be primary.

AI Use Disclosure

During the preparation of this work, the author used AI tools exclusively for stylistic and technical editing to improve readability. After using these tools, the output was carefully reviewed, edited, and verified by the author, who takes full responsibility for the conceptual architecture, analysis, and final content of the paper.

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